

Jayesh Dethe

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LinkedIn | GitHub | LeetCode

TECHNICAL SKILLS

- **Languages:** C, C++, Python, JavaScript
- **Backend & Web:** Flask, REST APIs, MongoDB, HTML, CSS, Chart.js
- **Machine Learning:** XGBoost, Feature Engineering, Cross Validation, SHAP
- **Deep Learning & GenAI:** Vision Transformers (ViT), Diffusion Models, Multimodal AI, Agentic AI
- **Computer Vision & 3D:** OpenCV, Open3D, MiDaS, ControlNet, TripoSR, Marching Cubes
- **Tools:** Git, GitHub, Render, Streamlit, LangGraph

WORK EXPERIENCE

National Institute of Technology, Tiruchirappalli

May 2025 – Jun 2025

Research Intern – 3D Vision

- Architected a volumetric fusion pipeline to reconstruct 3D geometry from sparse orthographic sketches using **OpenCV** and **ControlNet**.
- Integrated **Monocular Depth Estimation (MiDaS)** and designed normal-based voxel inflation algorithms to resolve depth ambiguities in 2D inputs.
- Implemented **voxel carving** strategies to generate structured voxel grids and produced watertight meshes using **Marching Cubes**.
- Applied **Taubin smoothing** to eliminate surface noise while preserving structural details, validated via rotating voxel visualizations.

PROJECTS

WonderWise – Agentic AI Travel Planner

Sep 2025

- Built a multi-agent system using **LangGraph** to decompose travel planning into research, recommendation, and itinerary agents.
- Integrated **Serper API** for real-time factual grounding and **Groq-hosted LLMs** to minimize inference latency.
- Implemented shared state management between agents to ensure consistent context during complex itinerary generation.

AI-Powered Text-to-3D Model Generation

Jul 2025

- Built an end-to-end generative pipeline converting natural-language prompts into 3D-printer-ready meshes.
- Orchestrated **SDXL Turbo** for rapid text-to-image synthesis and **U-2-Net** for precise background segmentation.
- Engineered a reconstruction module using **TripoSR**, leveraging **Vision Transformer (ViT)** features and **tri-plane representations** to predict implicit occupancy fields.
- Deployed the workflow via **Streamlit**, enabling real-time inference and mesh generation from raw text inputs.

Health Tracker Web App

Apr 2025

- Developed a full-stack health analytics platform to log metrics (BMI, BP, Sleep) with secure session-based authentication.
- Designed RESTful APIs using **Flask** and **MongoDB** and visualized health trends using interactive **Chart.js** dashboards.

Diabetes Risk Prediction System

Feb 2025

- Trained an **XGBoost** classifier achieving **93% accuracy** and **0.87 F1-score** on structured medical data.
- Implemented Z-score outlier removal and used **SHAP** values to interpret feature importance for medical risk factors.

EDUCATION

National Institute of Technology, Tiruchirappalli

2024 – 2026

M.Sc. Computer Science

CGPA: **8.70/10**

Ramnarain Ruia Autonomous College, Mumbai

2021 – 2024

B.Sc. Statistics

CGPA: **9.50/10**

ACHIEVEMENTS

- Recipient of National Scholarship Portal (NSP) Scholarship for postgraduate studies.
- Solved 200+ DSA problems on LeetCode and GeeksForGeeks.